MASH 1-1 ΔΣ Modulator

QUICK REVIEW SHEET • THEORY + DESIGN CHECKS

CORE IDEA

• MASH 1-1 = two cascaded first-order ΔΣ stages + digital noise cancellation.

• Each stage is first order; the combined ideal output has second-order noise shaping.

• Quantization noise is redistributed in frequency, not eliminated.

KEY DEFINITIONS

• Quantization error: e[n] = x[n] − y[n] (state the sign convention).

• Integrator: v[n] = v[n−1] + u[n]. In the loop, u[n] = input − feedback DAC.

• STF: signal transfer function. NTF: quantization-noise transfer function.

• OSR = f■/(2B) for a low-pass signal band of one-sided bandwidth B.

FIRST-ORDER STAGE

• Canonical model: Y■(z) = X(z) + (1 − z■⁻¹)E■(z).

• STF■(z) = 1; NTF■(z) = 1 − z■⁻¹.

• NTF has a zero at DC → low-frequency quantization noise is suppressed.

• Near DC, |NTF■| ≈ ω → about 20 dB/decade amplitude slope.

MASH 1-1 CANCELLATION — MEMORIZE THE LOGIC

• Stage 1: Y■ = X + (1 − z■⁻¹)E■.

• Stage 2 residue model: Y■ = E■ + (1 − z■⁻¹)E■.

• Combiner: Y = Y■ − (1 − z■⁻¹)Y■.

• Substitute and cancel E■: Y = X − (1 − z■⁻¹)²E■.

• Overall STF = 1; overall NTF = −(1 − z■⁻¹)². Minus sign does not change noise magnitude/order.

• Implementation rule: derive delays/gains from your actual stage equations; a one-sample mismatch breaks cancellation.

SECOND-ORDER NOISE SHAPING

• NTF■(z) = (1 − z■⁻¹)² = 1 − 2z■⁻¹ + z■⁻².

• Two zeros at DC; near DC, |NTF■| ≈ ω² → about 40 dB/decade amplitude slope.

• Oversampling creates out-of-band space for shaped noise; low-pass filtering + decimation removes much of it.

1.5-BIT QUANTIZER

• Usually means a 3-level quantizer: –Vref, 0, +Vref (normalized: –1, 0, +1).

• Three states commonly use 2 digital bits, with one code unused.

• Typically needs two thresholds/comparators; feedback DAC must match the three levels.

• Not every MASH 1-1 uses a 1.5-bit quantizer. Confirm which stage and exact thresholds/encoding.

VERIFICATION CHECKLIST

• 1) Test one first-order stage; output average should track input after scaling/transients.

• 2) Inspect PSD: first-order shaping.

• 3) Add Stage 2 and verify residue path.

• 4) Add cancellation and inspect PSD: second-order shaping.

• 5) Sweep input amplitude; check saturation, bounded integrator states, SNR/SNDR.

• 6) Test nonidealities only after the ideal model works.

COMMON EXAM / DESIGN TRAPS

• Two cascaded stages do not automatically create second-order shaping; cancellation is essential.

• Stage 2 must process a physical residue corresponding to Stage 1 error, with correct gain and delay.

• Do not mix equations with different delay conventions.

• SNR excludes distortion; SNDR includes distortion.

• Noise shaping is not noise elimination; the decimation filter suppresses out-of-band noise.